Practice Questions

1. With $S_1 = \{2, 3, 5, 7\}$, $S_2 = \{2, 4, 5, 8, 9\}$, and $U = \{1 : 10\}$, compute $\overline{S_1} \bigcup S_2$.

2. With $S_1 = \{2, 3, 5, 7\}$ and $S_2 = \{2, 4, 5, 8, 9\}$, compute $S_1 \times S_2$ and $S_2 \times S_1$.

3. Show that $S_1 = S_2$ if and only if $S_1 \cup S_2 = S_1 \cap S_2$.

4. If S_1 and S_2 are finite sets, show that $|S_1 \times S_2| = |S_1||S_2|$.

5. Prove each of the following.
   (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
   (c) $A \cap (A \cup B) = A$
   (d) $A \cup (A \cap B) = A$
   (e) $A - (B \cap C) = (A - B) \cup (A - C)$

6. Show that:
   $$(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$$

7. Without using truth table:
   Give an indirect proof of
   $$(\neg Q,\ P \Rightarrow Q,\ P \vee S) \Rightarrow S$$

8. Construct the truth table for the following:
   (a) $(P \vee Q) \Rightarrow ((P \vee R) \Rightarrow (R \vee Q))$
   (b) $(P \vee (Q \Rightarrow R)) \Leftrightarrow ((P \vee \neg R) \Rightarrow Q)$

9.

Prove the following equivalences:

(a) $(\neg P \Rightarrow (\neg P \Rightarrow (\neg P \wedge Q))) \equiv P \vee Q$

(b) $P \equiv (P \vee Q) \wedge (P \vee \neg Q)$

(c) $\neg (P \Leftrightarrow Q) \equiv (P \wedge \neg Q) \vee (\neg P \wedge Q)$

10.

Find grammars for $\Sigma = \{a, b\}$ that generate the sets of

(a) all strings with exactly two a's.

(b) all strings with at least two a's.

(c) all strings with no more than three a's.

(d) all strings with at least three a's.

(e) all strings that start with a and end with b.

(f) all strings with an even number of b's.

11.

What language does the grammar with these productions generate?

$$S \rightarrow Aa,$$
$$A \rightarrow B,$$
$$B \rightarrow Aa.$$

12.

Find a grammar for the language $L = \{a^n, \text{where n is even}\}$.

13.

Find a grammar for the language $L = \{a^n, \text{where } n \text{ is even and } n > 3\}$.

14.

Find grammars for $\Sigma = \{a, b\}$ that generate the sets of

(a) all strings with exactly two a's.

(b) all strings with at least two a's.

(c) all strings with no more than three a's.

(d) all strings with at least three a's.

(e) all strings that start with a and end with b.

(f) all strings with an even number of b's.

15.

Find three strings in the language generated by

$$S \rightarrow aSb|bSa|a.$$